



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
2025/2026 University Examination
FIRST SEMESTER OF THIRD YEAR,
BACHELOR OF SCIENCE IN MECHATRONIC ENGINEERING
EMT 3203: SOLID AND STRUCTURAL MECHANICS II

DATE: 9TH February 2026

TIME: 8.30AM-10.30AM

DURATION: 2 hours

1. This examination contains 70 marks in total.
2. Question 1 is compulsory.
3. Attempt one question in SECTION A and one question in SECTION B.

Question 1: Compulsory

[30 marks]

- (a) State three critical considerations when analyzing a crane beam supporting an engine during maintenance operations. [3 marks]
- (b) A cantilever beam of length $L = 2.0$ m carries a uniformly distributed load of $w = 100$ N/m. Calculate:
 - (i) The support reaction force [2 marks]
 - (ii) The maximum shear force and its location [1 marks]
 - (iii) The maximum bending moment and its location [2 marks]
- (c) Explain why the neutral axis in a beam under pure bending experiences zero stress.
What happens to stress distribution above and below the neutral axis? [4 marks]
- (d) A steel I-beam has the following properties: $I = 6.5 \times 10^6$ mm⁴, $c = 80$ mm. If the beam experiences a maximum bending moment of 4.5 kN·m, calculate:
 - (i) The section modulus S [1 marks]
 - (ii) The maximum bending stress [2 marks]
 - (iii) The safety factor if the yield strength is 250 MPa [2 marks]
- (e) List three factors that affect beam deflection and explain how each influences the magnitude of deflection. [3 marks]

- (f) Compare the stiffness of three support configurations (cantilever, simply supported, and fixed-fixed) for a beam of length $L = 1.5$ m carrying a center point load $P = 600$ N. Given: $E = 200$ GPa, $I = 1.8 \times 10^{-7}$ m⁴. Calculate the maximum deflection for each configuration and rank them from most to least stiff. [5 marks]
- (g) A PCB in an electronics rack has the following specifications: Length 180 mm, Width 120 mm, Thickness 1.6 mm, Material FR4 ($E = 20$ GPa), Component weight distributed uniformly 10 N total, Support fixed-fixed at both ends. Calculate the maximum deflection and assess whether it meets the standard (deflection $< L/100$). [5 marks]

SECTION A: Attempt Only One Question (20 marks)

Question 2:

[20 marks]

An electric train pantograph arm maintains contact with overhead power lines with the following specifications: Pin joint at A ($x = 0$), Total arm length $L = 1200$ mm, Spring location 300 mm from pin joint A, Wire contact force $P_1 = 800$ N downward at tip, Hollow steel tube OD = 50 mm, wall thickness = 4 mm, $I = 2.45 \times 10^6$ mm⁴, $c = 25$ mm, Material High-strength steel ($\sigma_{\text{yield}} = 250$ MPa), Safety factor required 3.0.

- (a) Calculate required spring force using equilibrium [4 marks]
- (b) Construct shear force diagram showing all regions [5 marks]
- (c) Determine bending moment distribution and locate maximum moment [5 marks]
- (d) Calculate maximum bending stress at critical section [3 marks]
- (e) Evaluate safety factor and design adequacy for dynamic railway operations [3 marks]

Question 3:

[20 marks]

A printed circuit board (PCB) is mounted between two supports in an electronics rack with the following specifications: PCB dimensions $L = 200$ mm, $b = 100$ mm, $h = 1.6$ mm, Total component weight 12 N (uniformly distributed), UDL intensity $w = 60$ N/m, Material FR4 composite ($E = 20$ GPa), Support type Fixed-fixed at both ends (screwed down).

- (a) Calculate the second moment of area [2 marks]
- (b) Calculate maximum midspan deflection [5 marks]
- (c) Assess deflection against the standard (deflection $< L/100$) [5 marks]
- (d) Propose and analyze two stiffening design modifications [8 marks]

SECTION B: Attempt Only One Question (20 marks)

Question 4:**[20 marks]**

A lightweight quadcopter drone uses hollow carbon fiber arms to support the motors and propellers. Each arm is modeled as a cantilever beam, fixed at the drone body and free at the motor end, and is subjected to both rotor thrust and motor torque.

The arm is a hollow circular tube with an outer diameter of $D = 20$ mm, an inner diameter of $d = 16$ mm, and a length of $L = 300$ mm. Its section properties are $I = 4618 \text{ mm}^4$, $J = 9236 \text{ mm}^4$, and $c = 10$ mm. The material is a carbon fiber composite with a yield strength of $\sigma_{\text{yield}} = 400$ MPa, and a target safety factor of 3–5. Each arm carries a downward rotor thrust of $F = 50$ N at the free end and a motor torque of $T = 3 \text{ N}\cdot\text{m}$ acting along its axis.

- (a) Compute the bending moment and bending stress at the fixed end. **[6 marks]**
- (b) Compute the torsional shear stress due to motor torque at the fixed end. **[4 marks]**
- (c) Determine the von Mises equivalent stress combining bending and torsion effects. **[6 marks]**
- (d) Evaluate whether the arm meets the safety factor requirement under combined loading. **[4 marks]**

Question 5:**[20 marks]**

The main shaft of a horizontal-axis wind turbine transmits power from the rotor to the gearbox while supporting the rotor and shaft self-weight. It is a hollow AISI 1045 steel tube with outer diameter $D = 100$ mm, inner diameter $d = 80$ mm, and length $L = 2.0$ m, supported by a pinned bearing at A and a roller bearing at B, with the rotor overhanging beyond bearing B.

The shaft carries a point load of $P_1 = 1200$ N at the rotor tip, a uniformly distributed load of $W = 300$ N/m along its length, and a constant torque of $T = 5000 \text{ N}\cdot\text{m}$. Its section properties are $I = 2.896 \times 10^6 \text{ mm}^4$, $J = 5.792 \times 10^6 \text{ mm}^4$, and $c = 50$ mm. The material yield strength is $\sigma_{\text{yield}} = 310$ MPa, with a target safety factor of 2.5–4.0.

- (a) Calculate the support reactions at bearings A and B. **[4 marks]**
- (b) Determine the location and magnitude of maximum bending moment. **[6 marks]**
- (c) Compute bending and torsional stresses at the critical section. **[6 marks]**
- (d) Assess the von Mises stress and evaluate the safety factor against yielding. **[4 marks]**

End of Questions